UCS615 Deep Learning

Recursion Cellular Image Classification

Project Overview

- **Task**: Classify siRNA (RNA interference) from 6-channel fluorescence microscopy images
- **Dataset**: RxRx1 (Kaggle competition)
- **Classes**: 1,108 different siRNA treatments
- **Challenge**: High intra-class variability due to batch effects

`Multi-channel Input`

`1108 Classes`

`4 Cell Types`

Approach

Models Tested

- ResNet-50 (6-channel adaptation)
- DenseNet-121 (dense connectivity)
- EfficientNet-B0 (compound scaling)
- ConvNeXt V2 Nano (modern convolutions)
- Vision Transformer ViT-Base/16
- Chaotic CNN (Skew-Tent, Logistic, Sine maps)
- ResNeXt-50 (32×4d)

Training Setup

- **Optimizer**: AdamW (lr=3e-4, wd=1e-4)
- **Scheduler**: Cosine Annealing
- **Image Size**: 320×320
- **Batch Size**: 32

What Worked

Best Results

Model	Val Accuracy	Cell Type
ResNeXt-50	55.07%	All 4 types
DenseNet-121	46.72%	HUVEC
EfficientNet-B0	44.85%	All 4 types
ResNet-50	37.49%	HUVEC

ResNeXt-50 Winner

DenseNet Strong

What Didn't Work

- **ConvNeXt V2 Nano**: 0.11% — failed to learn
- **ViT-Base/16**: 0.11% — no inductive biases

Why They Failed

- No proper pre-trained weights for 6-channel input
- Transfer learning from ImageNet didn't adapt
- Need domain-specific pre-training

ConvNeXt V2

Vision Transformer

Chaotic CNN Variants

Tested Maps

- Skew-Tent Map → 30.56%
- Logistic Map (μ=4) → 32.22%
- Sine Map → 33.72%

Insight

Chaotic dynamics provided regularization but didn't outperform standard DenseNet-121

Key Takeaways

1. **ResNeXt-50** excels on large-scale biomedical classification
2. **DenseNet-121** superior to ResNet-50 due to feature reuse
3. Transfer learning needs proper 6-channel adaptation
4. ViT/ConvNeXt require domain-specific pre-training
5. Aggregated residuals > deeper networks

Conclusion

- **Best Model**: ResNeXt-50 (55.07% val accuracy)
- **Runner-up**: DenseNet-121 (46.72%)
- Traditional CNNs still outperform transformers on microscopy
- Multi-cell-type training improves generalization